

Dataset	Models	RM Type					Length	PPL
		LLaMA	BLOOMZ	Ours-LLaMA	Ours-BLOOMZ	GPT-J		
HH-RLHF	BLOOMZ-SFT	0.986	-1.008	1.098	0.285	0.162	502	10.09
	Vanilla-PPO-BLOOMZ	1,082	-1.161	1.183	-0.422	0.174	513	10.06
	Ours-PPO-BLOOMZ	1.867	-1.162	1.313	-0.327	0.167	507	9.13
	LLaMA-SFT	1.386	-1.179	1.854	0.354	0.174	468	10.16
	Vanilla-PPO-LLaMA	1.580	-0.997	2.106	-0.200	0.196	503	9.81
	Ours-PPO-LLaMA	1.474	-1.062	1.975	-0.256	0.172	485	8.92
	Alpaca-SFT	1.557	-1.008	2.152	0.413	0.191	513	10.11
	Vanilla-PPO-Alpaca	2.052	-0.623	2.882	0.152	0.241	689	9.67
	Ours-PPO-Alpaca	2.041	-0.587	2.850	0.150	0.221	586	8.82
	Biasd-PPO-Alpaca	1.948	-0.604	2.930	0.025	0.148	684	9.91
rm-static	BLOOMZ-SFT	1.463	0.101	1.315	0.293	0.164	494	10.03
	Vanilla-PPO-BLOOMZ	1.599	0.242	1.480	0.421	0.241	517	9.98
	Ours-PPO-BLOOMZ	1.688	0.324	1.603	0.487	0.239	501	8.84
	LLaMA-SFT	2.026	0.365	2.367	0.530	0.182	473	9.53
	Vanilla-PPO-LLaMA	2.101	0.463	2.480	0.599	0.262	513	9.30
	Ours-PPO-LLaMA	2.045	0.430	2.464	0.585	0.249	489	8.89
	Alpaca-SFT	2.224	0.597	2.807	0.736	0.193	532	9.53
	Vanilla-PPO-Alpaca	2.563	0.931	3.414	0.931	0.291	721	9.03
	Ours-PPO-Alpaca	2.587	0.919	3.428	0.968	0.289	617	8.80
	Biasd-PPO-Alpaca	2.848	1.260	3.845	1.232	0.152	713	9.57

Table 1: Main result for our proposed framework, we trained multiple models following RLHF pipeline. And we use five different Reward Model to comprehensively evaluate the sentences against different settings, which are generated from model-generated 4608 and 2304 prompts extracted from hh-rlhf and rm-static respectively.

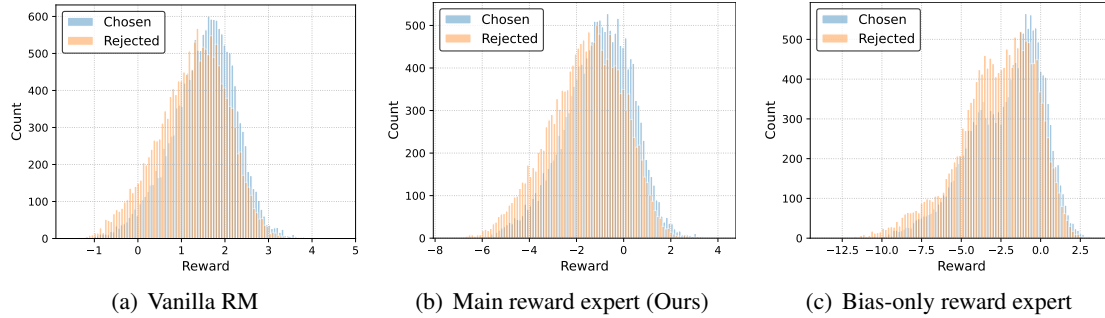


Figure 5: Distribution of reward scores for chosen and rejected responses. The results demonstrate that our proposed method increases the differentiation between chosen and rejected responses. In contrast, the bias-only expert exhibits limited generalization ability, primarily relying on length bias.

Models	RM %	Bias-only %
Vanilla-BLOOMZ	66.94	—
w/o PoE	67.11	66.73
w/o Input Noise	67.53	66.58

Table 2: Ablation study on HH dataset, RM Accuracy and bias-only RM for BLOOMZ models. By employing a combination of PoE and input noise perturbation on enhancing the generalization capability of the reward model. The performance on the test set significantly improves as a result of these techniques

Metrics We evaluate the effectiveness of different methods in our experiments using perplexity (gpt2-medium), average reward score, and human

evaluators. For human evaluation, annotators compared two randomly selected responses and provided comparison results (win/lose/tie), allowing us to gain insights into subjective judgments of response quality.

6.2 True Reward Improvement

As the previous analysis, addressing the risk of reward hacking resulting from reward model overoptimization (Gao et al., 2022) is crucial. To ensure a more robust evaluation of our method, we adopted a comprehensive approach that combines both automatic and human evaluation. Table 1 illustrates the average generated sequence length and reward scores of our method in comparison to

the baselines on the test set. The results clearly demonstrate that our approach achieves higher reward scores while simultaneously reducing the average output length compared to the RL model utilizing the vanilla reward model. This finding not only confirms the effectiveness of our proposed method but also establishes a foundation for generating outputs that better align with human preferences.

In addition, we conducted a Spearman/Pearson analysis to further validate the effectiveness of our method. For more detailed information, please refer to Appendix 4.

	Opponent	Human	AlpacaFarm	GPT-4
Ours	Answer _{Chosen}	67.69	74.42	75.32
Ours	SFT	54.23	59.42	56.42
Ours	PPO	57.47	61.43	59.56

Table 3: Win rates of our proposed method against various baseline approaches in the HH-RLHF dataset, utilizing the Alpaca. Evaluations include human annotators, AlpacaFarm and GPT-4. Answer_{Chosen} is the human preferred response in HH-RLHF.

6.3 Wining Rate

In this section of the experimental analysis, we present the win rate of our method compared to other approaches. We provide results from both manual evaluations, GPT4 and the automated evaluation platform, and AlpacaFarm. During the pairwise comparisons of the model results, a win is assigned 1 point, a tie is assigned 0.5 points, and a loss is assigned 0 points. It is evident that our method achieves an average win rate of more that 50% compared to traditional RLHF methods. This validates the effectiveness of our proposed method in generating outputs that are more informative and concise.

6.4 Ablation Study

Table 2 presents the results of ablation analysis on various components of our method. By utilizing the PoE technique and input perturbation, we observe an improvement in the accuracy of the RM. This finding demonstrates that our method can better identify human intent, leading to enhanced performance.

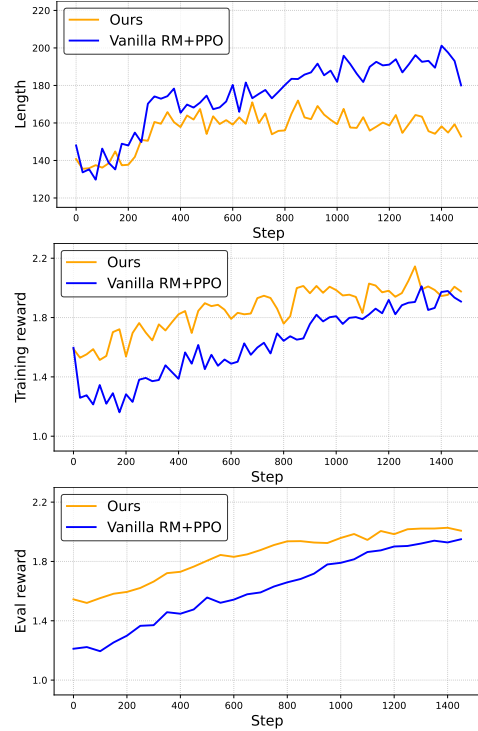


Figure 6: Comparison of training curves between our proposed method and PPO using vanilla RM. The results show that our method achieves stable increases in reward values while ensuring consistent generation of output lengths.

7 Analysis and Discussion

In this section, we uncover valuable insights into the effectiveness and limitations of our approaches, paving the way for future advancements in conversational AI systems.

7.1 Leaning Curve

Figures 6 present the variation in generated sequence length during the training process of our method and the vanilla RM for PPO, along with the output reward scores during training and on the test set. From the Figures, it can be observed that the model trained with the vanilla RM continuously increases its output length throughout the training process, whereas our method achieves a stable output length after a slight increase. Both methods enhance the output’s reward score, but our approach yields a more concise output while containing more informative content.

7.2 Distribution of Reward Scores

Figure 5 illustrates the distribution of reward scores for the chosen and rejected data on the validation set during the training of our method compared to the vanilla reward model. It is evident that

our model exhibits better discernment between the chosen and rejected data. This improved performance can be attributed to our approach’s ability to avoid excessive learning bias towards sequence length. By mitigating the length bias, the main expert of our model can focus more on understanding genuine human intent, leading to enhanced generalization capabilities.

8 Conclusion

In this study, we investigate the issue of length bias in NLP and propose a PoE-based method to mitigate this bias. Our work sheds light on the challenges associated with reward modeling, emphasizing that it is not a straightforward task and various difficulties may impede the models from capturing the true human intent. By addressing the problem of length bias, our research highlights the importance of developing techniques that enable models to learn and generate responses aligned with genuine human intentions.

Further research and advancements in this area are necessary to overcome these obstacles and enhance the performance and reliability of NLP models in real-world applications.

Limitations

In this study, we propose a simple and effective method to mitigate length bias during the Reinforcement Learning from Human Feedback stage. However, it is important to note that our method can only alleviate length bias to some extent and may not completely eliminate it. Furthermore, the validation of our method’s effectiveness was conducted on two RLHF datasets. It is worth mentioning that collecting RLHF data is a challenging task, and the question of whether this phenomenon exists on larger datasets remains uncertain. Evaluating the performance of general dialogue models poses a difficulty. Therefore, during the human evaluation phase, we selected only a limited number of evaluation samples to assess the effectiveness of our approach.

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